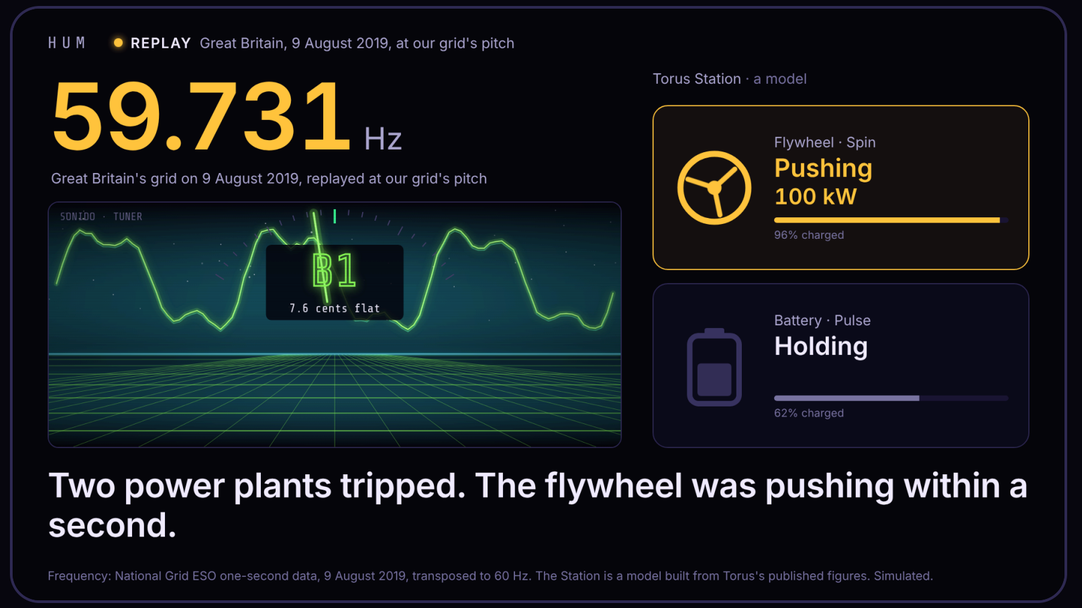
Hard Tech Week 2026 · Torus bounty · Nucleus × Startup State open call

HUM hears the power grid through a guitar, and a model of a Torus Station answers it.

The software that decides is written in Mentl, a programming language I built. Its compiler proves that this software can't reach the internet while it decides.

Morgan Espitia, Salt Lake City. I have built Mentl since March and my audio library, sonido, since January.



The live screens read the grid from a room's wiring, except the replayed 2019 event.

ampactor.dev/hum

Data: grid frequency, NESO open data at 1 s, 9 Aug 2019, moved to 60 Hz · rush hour, EIA-930 hourly demand, PacifiCorp East, 19–26 Oct 2026 · Station: a simulated model of Torus's published figures

How to read this demo

The power grid's frequency shows whether it has enough power. A guitar can hear it.

The grid	The whole Western grid, from Alberta to Baja, runs at one frequency: 60 hertz. When demand outruns supply, the frequency falls.
The ear	Every power adapter hums at that frequency. A guitar pickup resting on one hears it, and my audio library, sonido, measures it to a thousandth of a hertz.
The Station	A model of a Torus Station: a flywheel for the first seconds (100 kW for six minutes) and a battery for the hours (300 kW for four hours). Simulated.
The software	Decides what the Station does. It is written in Mentl, a language I built, whose compiler proves the decision can't reach the internet.
Next	The grid's hum in a room's wiring. Then tomorrow's rush hour, a real grid failure from 2019, and a blackout.

Station figures: Torus's published specifications, modeled and simulated. Frequency: read from a room's wiring.

AI now helps write the software that runs machines. No mainstream language can prove what that software will never do.

Batteries, flywheels and flight controls run on code that people test and review, and then trust.

Mentl proves what software can never do.

You write a rule into the code, such as "this never touches the internet" or "this never writes a file".

The compiler checks the rule through every function the code calls. **Break it, and the program won't build.**

Seven months and 2,600 commits since March. The compiler is written in Mentl and compiles itself. Open source: github.com/ampactor-labs/mentl

The compiler's answer about the controller in this demo

```
$ mentl why examples/grid/station.mn decide Network
decide never uses Network: nothing in decide, and
nothing it calls, performs an operation of Network

$ mentl why examples/grid/station.mn decide Alloc
decide never uses Alloc: nothing in decide, and
nothing it calls, performs an operation of Alloc
```


Counts: the git history of github.com/ampactor-labs/mentl. The answers: pitch/why.txt, captured from the compiler on 28 Oct 2026.

What Mentl code looks like

Five operators draw the shape of a program. The compiler checks each rule along that shape.

`fn read(hz) = hz |> fast`	`|>`	pipe: one stage feeds the next
`fn fork(hz) = (hz) <| (fast, slow)`	`<|`	split: one value goes to two branches
`fn pair(a, b) = (a |> fast) >< (b |> slow)`	`><`	parallel: two branches run side by side
`fn trend(hz) = ((prev) => prev + hz) <~ Delay(1)`	`<~`	feedback: each step sees the step before
`fn act(hz) with !Network = spin(hz) ~> flywheel`	`~>`	handler: names who answers the request

The last line carries a rule, `with !Network`: if anything `act` calls could reach the network, the compiler refuses to build it. Four of the operators come from Faust, the audio language; `~>` is Mentl's own.

These lines pass mentl check as written, 28 Oct 2026. The language: github.com/ampactor-labs/mentl

Everything you just saw was built since Monday at 4 PM, on two tools I built before it.

Before Mentl, the language: 2,610 commits since March 15. sonido, the audio library: 461 commits since January 25.

This week All of HUM: the ear, the controller, the director, the page and this show. In sonido, 10 commits, among them the grid ear and the pedal. In Mentl, 27.

Check it Every commit, with its time: github.com/ampactor-labs/hum, printed by `./ledger.sh`

Counts: each repository's git history since Monday 26 Oct 2026, 4:00 PM MDT, printed in pitch/ledger.txt.

Who needs it in Utah

TORUS · SALT LAKE CITY

Builds flywheel-and-battery Stations that steady the grid. The software inside each Station is where a proof would matter first. Torus is judging this demo.

HILL AFB · 309TH SOFTWARE ENGINEERING GROUP

Maintains flight software for the F-16, the F-35 and the missile fleet, where every claim about the software needs evidence.

First conversation I'm asking for: Torus engineering, this month.
First hires: a controls engineer and a compiler engineer, in Salt Lake City.

What $10,000 unlocks

$10,000 puts this controller, proven unable to reach the internet, beside a real battery in Utah within ninety days.

Day 30 Form the Utah company. Run the controller on a battery inverter on my bench.

Day 60 Run it beside a real battery at one Utah site.

Day 90 A pilot agreement with that site, and an SBIR proposal with Nucleus Grow.

$7,000 for the inverter and battery bench. **$3,000** to form the company, register for federal grants and install at the site.

Three asks. A Utah operator who will host it. An introduction to Hill AFB's 309th. A co-founder who knows power electronics.

Thank you, Torus and Nucleus

Replay this demo on your phone.



`ampactor.dev/hum`

Morgan Espitia · sonofagoose@pm.me

The demo: github.com/ampactor-labs/hum
The language: github.com/ampactor-labs/mentl

Three asks: a Utah operator who will host the pilot, an introduction to Hill AFB's 309th, and a co-founder who knows power electronics.